

LINUX COMMAND REFERENCE

1. File and Directory Management

- `ls` – List directory contents
- `cd` – Change directory
- `pwd` – Print working directory
- `cp` – Copy files and directories
- `mv` – Move or rename files and directories
- `rm` – Remove files or directories
- `mkdir` – Make directories
- `rmdir` – Remove empty directories
- `touch` – Change file timestamps or create empty files
- `find` – Search for files in a directory hierarchy
- `locate` – Find files by name
- `tree` – Display directories in a tree-like format
- `chmod` – Change file permissions
- `chown` – Change file owner and group
- `chgrp` – Change group ownership
- `stat` – Display file or file system status

2. File Viewing and Editing

- `cat` – Concatenate and display file content
- `tac` – Concatenate and display file content in reverse
- `more` – View file content interactively (page by page)
- `less` – View file content interactively (scrollable)
- `head` – Output the first part of a file
- `tail` – Output the last part of a file
- `nano` – Text editor (terminal-based)
- `vim / vi` – Advanced text editors
- `emacs` – Text editor
- `grep` – Search text using patterns
- `sed` – Stream editor for filtering and transforming text
- `awk` – Pattern scanning and processing language

- **cut** – Remove sections from each line of files
- **sort** – Sort lines of text files
- **uniq** – Report or omit repeated lines

3. Process Management

- **ps** – Report a snapshot of current processes
- **top** – Display Linux tasks
- **htop** – Interactive process viewer (advanced top)
- **kill** – Send a signal to a process, typically to terminate
- **killall** – Terminate processes by name
- **bg** – Resume a suspended job in the background
- **fg** – Bring a job to the foreground
- **jobs** – List active jobs
- **nice** – Run a program with modified scheduling priority
- **renice** – Alter priority of running processes
- **uptime** – Show how long the system has been running
- **time** – Measure program running time

4. Disk Management

- **df** – Report file system disk space usage
- **du** – Estimate file space usage
- **fdisk** – Partition table manipulator for Linux
- **lsblk** – List information about block devices
- **mount** – Mount a file system
- **umount** – Unmount a file system
- **parted** – A partition manipulation program
- **mkfs** – Create a file system
- **fsck** – File system consistency check and repair
- **blkid** – Locate/print block device attributes

5. Networking

- **ifconfig** – Configure network interfaces
- **ip** – Show/manipulate routing, devices, and tunnels
- **ping** – Send ICMP Echo requests to network hosts
- **netstat** – Network statistics

- **ss** – Socket statistics (faster than netstat)
- **traceroute** – Trace the route packets take to a network host
- **nslookup** – Query Internet name servers interactively
- **dig** – DNS lookup utility
- **wget** – Non-interactive network downloader
- **curl** – Transfer data with URLs
- **scp** – Secure copy files between hosts
- **ssh** – Secure shell for remote login
- **ftp** – File Transfer Protocol client

6. User and Group Management

- **useradd** – Add a user to the system
- **usermod** – Modify a user account
- **userdel** – Delete a user account
- **groupadd** – Add a group to the system
- **groupdel** – Delete a group
- **passwd** – Change user password
- **chage** – Change user password expiry information
- **whoami** – Print the current logged-in user
- **who** – Show who is logged in
- **w** – Show who is logged in and what they're doing
- **id** – Display user and group information
- **groups** – Show user's groups

7. System Information and Monitoring

- **uname** – Print system information
- **hostname** – Show or set the system's hostname
- **uptime** – How long the system has been running
- **dmesg** – Boot and system messages
- **free** – Display memory usage
- **top** – Display Linux tasks
- **vmstat** – Report virtual memory statistics
- **lscpu** – Display information about the CPU architecture
- **lsusb** – List USB devices
- **lspci** – List PCI devices

- **lshw** – List hardware configuration

8. Archiving and Compression

- **tar** – Archive files
 - **tar -czf archive.tar.gz /path/to/directory** – Compress files using gzip
 - **tar -xzf archive.tar.gz** – Extract gzipped tarball
 - **tar -cf archive.tar /path/to/directory** – Create a tarball
 - **tar -xf archive.tar** – Extract tarball
- **zip** – Package and compress files into a ZIP archive
- **unzip** – Extract files from a ZIP archive
- **gzip** – Compress files using the gzip algorithm
- **gunzip** – Decompress files compressed with gzip
- **bzip2** – Compress files using the bzip2 algorithm
- **bunzip2** – Decompress files compressed with bzip2
- **xz** – Compress files using the xz algorithm
- **unxz** – Decompress files compressed with xz

9. Package Management (Depends on Distribution)

Debian-based (e.g., Ubuntu)

- **apt-get** – APT package handling utility
 - **apt-get install <package>** – Install a package
 - **apt-get update** – Update package list
 - **apt-get upgrade** – Upgrade installed packages
 - **apt-get remove <package>** – Remove a package
- **apt-cache** – Query APT cache
 - **apt-cache search <package>** – Search for a package
 - **apt-cache show <package>** – Show package details

Red Hat-based (e.g., CentOS, Fedora)

- **yum** – Package manager for RPM-based systems
 - **yum install <package>** – Install a package
 - **yum update** – Update installed packages
 - **yum remove <package>** – Remove a package

- **dnf** – Next-generation package manager (Fedora, CentOS 8+)
 - **dnf install <package>** – Install a package
 - **dnf update** – Update installed packages
 - **dnf remove <package>** – Remove a package

General Commands

- **rpm** – RPM package manager
 - **rpm -i <package.rpm>** – Install an RPM package
 - **rpm -e <package>** – Remove an RPM package
- **dpkg** – Debian package manager
 - **dpkg -i <package.deb>** – Install a Debian package
 - **dpkg -r <package>** – Remove a Debian package

System Services and Daemon Management

- **systemctl** – Control the systemd system and service manager
 - **systemctl start <service>** – Start a service
 - **systemctl stop <service>** – Stop a service
 - **systemctl restart <service>** – Restart a service
 - **systemctl enable <service>** – Enable a service to start on boot
 - **systemctl disable <service>** – Disable a service from starting on boot
 - **systemctl status <service>** – Check service status
- **service** – Older service management command (used in nonsystemd systems)
 - **service <service> start** – Start a service
 - **service <service> stop** – Stop a service
 - **service <service> restart** – Restart a service
 - **service <service> status** – Check service status

10. Scheduling Tasks

- **cron** – Daemon for running scheduled commands
 - **crontab -e** – Edit cron jobs for the current user
 - **crontab -l** – List the current user's cron jobs
 - **crontab -r** – Remove the current user's cron jobs
- **at** – Run commands at a specified time
 - **at 09:00** – Schedule a command to run at 09:00 AM

- **batch** – Run commands when the system load is low
- **sleep** – Delay for a specified time
 - **sleep 5s** – Sleep for 5 seconds

11. File Permissions and Security

- **chmod** – Change file permissions
- **chown** – Change file owner and group
- **chgrp** – Change the group ownership of a file
- **umask** – Set default permissions for new files
- **setfacl** – Set file access control lists (ACL)
- **getfacl** – Get file access control lists (ACL)
- **sudo** – Execute a command as another user (usually root)
- **visudo** – Edit the sudoers file safely
- **passwd** – Change a user's password
- **sudoers** – Manage sudo access for users
- **gpasswd** – Administer group password
- **ss** – Display socket statistics (for secure network connections)

12. System Backup and Restore

- **rsync** – Remote file and directory synchronization
 - **rsync -avz source/ destination/** – Synchronize files
 - **rsync -avz -e ssh source/ user@remote:/destination/** – Sync over SSH
- **cpio** – Copy files to and from archives
- **dd** – Low-level copying and backup of entire filesystems
 - **dd if=/dev/sda of=/path/to/backup.img** – Backup a disk/partition
 - **dd if=/path/to/backup.img of=/dev/sda** – Restore a disk/partition

13. System Diagnostics and Troubleshooting

- **dmesg** – Print the kernel ring buffer messages (system boot and hardware-related messages)
- **journalctl** – Query and view logs from systemd's journal
- **strace** – Trace system calls and signals
 - **strace <command>** – Trace a command's system calls
- **lsof** – List open files (useful for debugging)
 - **lsof <file>** – Show processes using a specific file

- **vmstat** – Report virtual memory statistics
- **iostat** – Report CPU and I/O statistics
- **mpstat** – Report CPU usage statistics
- **pidstat** – Report statistics by process
- **free** – Display memory usage
- **uptime** – How long the system has been running
- **watch** – Execute a program periodically, showing output
 - **watch -n 1 free** – Watch memory usage every second
- **lshw** – List hardware configuration
- **htop** – Interactive process viewer (better than top)
- **netstat** – Network statistics (deprecated in favor of ss)
- **ss** – Show socket statistics (more efficient than netstat)

14. Networking & Remote Management

- **ifconfig** – Configure network interfaces (older command, replaced by ip)
- **ip** – A more modern alternative for managing network interfaces and routing
 - **ip addr** – Show IP addresses
 - **ip link** – Show or manipulate network interfaces
 - **ip route** – Show or manipulate routing tables
- **ss** – Display socket statistics (useful for diagnosing network issues)
- **nmap** – Network exploration tool (can be used for security auditing)
- **telnet** – User interface to the TELNET protocol (less common nowadays)
- **nc (Netcat)** – Network utility for reading and writing from network connections
 - **nc -l -p 1234** – Listen on port 1234
 - **nc <host> <port>** – Connect to a host and port
- **iptables** – Administration tool for IPv4 packet filtering and NAT (Network Address Translation)
- **firewalld** – Frontend for managing firewall rules (used in some distros like Fedora and CentOS)
- **ufw** – Uncomplicated firewall (front-end for iptables)
 - **ufw enable** – Enable firewall
 - **ufw allow <port>** – Allow traffic on a specific port
- **tcpdump** – Command-line packet analyzer
- **curl** – Transfer data from or to a server using various protocols (HTTP, FTP, etc.)
- **wget** – Download files from the web via HTTP, HTTPS, FTP

- **scp** – Secure copy over SSH (used to copy files between systems)
 - `scp file.txt user@remote:/path/to/destination/` – Copy file to remote server
- **rsync** – Remote file and directory synchronization (often used for backups)
 - `rsync -avz /local/path/ remote:/remote/path/` – Sync directories

15. Text Processing Utilities

- **grep** – Search for patterns within files
 - `grep 'pattern' file.txt` – Search for a pattern in a file
 - `grep -r 'pattern' /dir/` – Recursively search for a pattern
- **sed** – Stream editor for filtering and transforming text
 - `sed 's/old/new/g' file.txt` – Replace old with new globally
- **awk** – A powerful text processing language
 - `awk '{print $1}' file.txt` – Print the first column of each line in a file
- **cut** – Remove sections from each line of a file
 - `cut -d ':' -f 1 /etc/passwd` – Print the first field of each line, delimited by ":"
- **sort** – Sort lines of text files
 - `sort file.txt` – Sort file content in ascending order
- **uniq** – Report or omit repeated lines in a file
 - `sort file.txt | uniq` – Sort and remove duplicate lines
- **tee** – Read from standard input and write to standard output and files
 - `echo "text" | tee file.txt` – Write to file and show output on screen
- **tr** – Translate or delete characters
 - `echo "hello" | tr 'a-z' 'A-Z'` – Convert lowercase to uppercase
- **paste** – Merge lines of files
 - `paste file1.txt file2.txt` – Combine lines of file1 and file2 side by side
- **wc** – Word, line, character, and byte count
 - `wc -l file.txt` – Count lines in a file
 - `wc -w file.txt` – Count words in a file

16. System Shutdown and Reboot

- **shutdown** – Shut down the system
 - `shutdown -h now` – Immediately shut down

- `shutdown -r now` – Reboot the system
- `shutdown -h +10` – Shut down after 10 minutes
- `reboot` – Reboot the system
- `halt` – Halt the system immediately (equivalent to turning off power)
- `poweroff` – Power off the system
- `init` – Change the runlevel (old-style system manager)
 - `init 0` – Shutdown
 - `init 6` – Reboot

17. File System Mounting and Management

- `mount` – Mount a file system
 - `mount /dev/sda1 /mnt` – Mount partition to a directory
- `umount` – Unmount a file system
 - `umount /mnt` – Unmount the file system mounted at /mnt
- `fstab` – File system table (configuration file for mounting file systems)
 - `/etc/fstab` – View and configure persistent mount points
- `blkid` – Display block device attributes
- `fsck` – Check and repair a file system
 - `fsck /dev/sda1` – Check and repair /dev/sda1

18. Filesystem Permissions and Security

- `chmod` – Change file permissions
 - `chmod 755 file.txt` – Give read, write, and execute permissions to owner, and read-execute permissions to others
- `chown` – Change file owner and group
 - `chown user:group file.txt` – Change owner and group of a file
- `chgrp` – Change group ownership of a file
 - `chgrp group file.txt` – Change the group of a file
- `umask` – Set default permissions for new files
 - `umask 022` – Set default permissions for newly created files to 755
- `setfacl` – Set access control lists (ACL) for file permissions
- `getfacl` – Get access control lists (ACL) for file permissions